

Predicting Stroke Risk Using Patient Health and Lifestyle Data

Data Analysis Report

Team CodeNexus

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1 Team Details

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2 Abstract

This report presents a stroke risk prediction study built on patient health and lifestyle data. The original dataset contained 9,722 records, but deduplication revealed 4,612 exact clinical duplicates that had artificially inflated the stroke positive class. After correction, the true dataset contains 5,110 unique patients with a stroke prevalence of 4.87%, consistent with published real world clinical rates.

The analysis combines exploratory data analysis, statistical hypothesis testing, feature engineering, and a leakage-safe machine learning pipeline. Five models are evaluated: Logistic Regression, Random Forest, XGBoost, Soft Voting Ensemble, and a Stacking Ensemble using out of fold meta learning. Under a clinically appropriate High Sensitivity operating mode, Logistic Regression achieves the highest recall (0.960), missing only 2 stroke patients out of 50 in the test set. The Stacking Ensemble achieves the best overall precision recall performance (AUC-PR = 0.2278). A key finding is that simpler regularised models can outperform gradient boosting on recall when the minority class is small, due to the bias variance tradeoff.

3 Introduction

3.1 Background and Motivation

Stroke is a major global health issue and a leading reason of death and long-term disability, affecting around 15 million people each year. Strokes generally results in severe complications such as paralysis, speech impairment, and cognitive decline. Early identification of high risk patients is crucial for preventing strokes and to reduce their impact. However, converting large volumes of medical data into accurate and actionable risk predictions remains a key challenge in healthcare analytics.

3.2 Problem Statement

This study addresses the challenge of predicting stroke events using a dataset of 9,722 patient records collected by a healthcare organisation. After removing duplicate records introduced through an artificial balancing procedure, the true dataset contains 5,110 unique patients with a stroke prevalence of 4.87%. The analysis addresses three core objectives:

- Develop a machine learning pipeline capable of accurately predicting stroke events in the presence of severe class imbalance
- Identify the most clinically influential risk factors contributing to stroke occurrence
- Generate evidence based insights to support preventive healthcare strategies and early screening protocols

3.3 Significance of the Study

A key contribution of this study is the identification and correction of an artificial dataset imbalance. The original 9,722 records contain 4,612 exact clinical duplicates concentrated entirely in the stroke-positive class, inflating a naturally rare event to a misleadingly balanced 50/50 distribution. Models trained on this artificially balanced data produce inflated performance metrics that do not reflect real world deployment conditions. This study works exclusively with the 5,110 deduplicated records, employing SMOTE oversampling, class weight adjustment, and threshold tuning to address the genuine imbalance in a methodologically sound manner.

4 Dataset and Preprocessing

4.1 Dataset Description

The dataset contains 17 variables covering demographics, medical history, lifestyle habits, and clinical measurements, with `stroke_event` as the binary target variable.

Variable	Type	Description	Modelling Status
<code>patient_id</code>	Identifier	Unique record identifier	Excluded
<code>gender</code>	Categorical	Patient gender (M / F)	Encoded (label)
<code>age</code>	Continuous	Age in years — strongest predictor ($r = 0.243$)	Retained
<code>age_group</code>	Categorical	Binned age: young / middle / senior	Excluded (redundant with age)
<code>has_hypertension</code>	Binary	Presence of hypertension (1 = Yes)	Retained
<code>has_heart_disease</code>	Binary	Presence of heart disease (1 = Yes)	Retained
<code>marital_status</code>	Binary	Marital status (1 = Married)	Retained
<code>employment_type</code>	Categorical	Work status: working / government / other	Encoded (label)
<code>residence</code>	Categorical	Urban or Rural area of residence	Encoded (label)
<code>glucose_level</code>	Continuous	Blood glucose level (mg/dL)	Retained
<code>high_glucose</code>	Binary	Binary threshold of <code>glucose_level</code>	Excluded (redundant)

Variable	Type	Description	Modelling Status
bmi_value	Continuous	Body Mass Index — 201 missing post-dedup	Imputed (grouped median)
bmi_category	Categorical	BMI classification	Excluded (redundant)
smoking_habit	Categorical	Smoking status (4 categories)	Encoded (label)
lifestyle_risk	Categorical	Composite lifestyle risk level	Excluded (leakage risk)
risk_score	Ordinal	Composite health risk score (0–2)	Excluded (leakage risk)
stroke_event	Binary (Target)	Stroke occurrence (1 = Yes, 0 = No)	Target variable

4.2 Data Integrity: Duplicate Detection and Removal

Original Data has 9,722 observations. After cleaning the duplicate data based on all clinical variables except the patient identifier variable, there remain 5,110 patients. This is important since all duplicate data occurred in only one class—stroke positive. If retained, this will amount to data leakage since the same observation would be used for both training and testing the model, resulting in inflated model metrics.

The corrected distribution after removing duplicates is as follows:

- **No Stroke:** 4,861 patients (95.13%)
- **Stroke:** 249 patients (4.87%)

This distribution aligns well with documented real-world prevalence of stroke ranging between 2-5%. The high stroke prevalence rate for **risk_score** = 2 at 84.5% and **lifestyle_risk** = **medium** at 62.6% suggests these composite variables may have been formed using stroke-related data. Both were excluded from all modelling.

4.3 Feature Engineering

Seven interaction and transformation features were engineered prior to the train test split, capturing clinically meaningful non linear relationships that raw features cannot represent independently:

Feature	Formula	Clinical Rationale
age_squared	age^2	Nonlinear acceleration of stroke risk with advancing age
is_senior	1 if age $\geq$ 55	Clinical threshold where stroke risk doubles per decade
age_x_hypertension	$\text{age} \times \text{hypertension}$	Compounding interaction of the two primary risk factors
age_x_heart_disease	$\text{age} \times \text{heart disease}$	Elevated risk in older cardiac patients

Feature	Formula	Clinical Rationale
<code>cvd_count</code>	<code>hypertension + heart_disease</code>	Cumulative cardiovascular disease burden (0–2)
<code>glucose_x_bmi</code>	<code>glucose × bmi</code>	Joint metabolic risk of hyperglycaemia and obesity
<code>log_glucose</code>	<code>log(glucose + 1)</code>	Normalises the right-skewed glucose distribution

4.4 Missing Value Imputation

After de-duplication, there are 201 records (3.93%) which contain null values in `bmi_value`. A grouped median strategy was used to impute the missing value such that the value was imputed using the median of the `bmi_value` for those who are of the same gender and age range (young if less than 40 years old, middle if between 40-60 years old, and senior if greater than 60 years old). This method is more biologically meaningful compared to the use of the global median.

The forward fill and backward fill strategies were considered but not adopted since they can be used only when the data are timeseries wherein there is temporal correlation among records. This is not true for patient data.

4.5 Leakage Prevention

Two categories of features were excluded from all modelling:

1. **Redundant derived features** that are direct functions of retained features: `age_group`, `bmi_category`, `high_glucose`
2. **Composite scores with high leakage risk**: `risk_score`, `lifestyle_risk`

These columns are either redundant or too closely tied to the target to be safe for honest prediction.

4.6 Categorical Encoding

All categorical variables were label encoded using scikit-learn's `LabelEncoder` fitted on training data only. The four encoded features are: `gender`, `employment_type`, `residence`, and `smoking_habit`. Standard scaling was not applied to tree-based models (Random Forest, XGBoost), which are invariant to feature scale. Logistic Regression used `class_weight='balanced'` as a built in mechanism to handle the imbalanced target.

5 Exploratory Data Analysis

5.1 Class Balance

The corrected dataset shows the true clinical imbalance: 4,861 non-stroke cases (95.13%) and 249 stroke cases (4.87%).

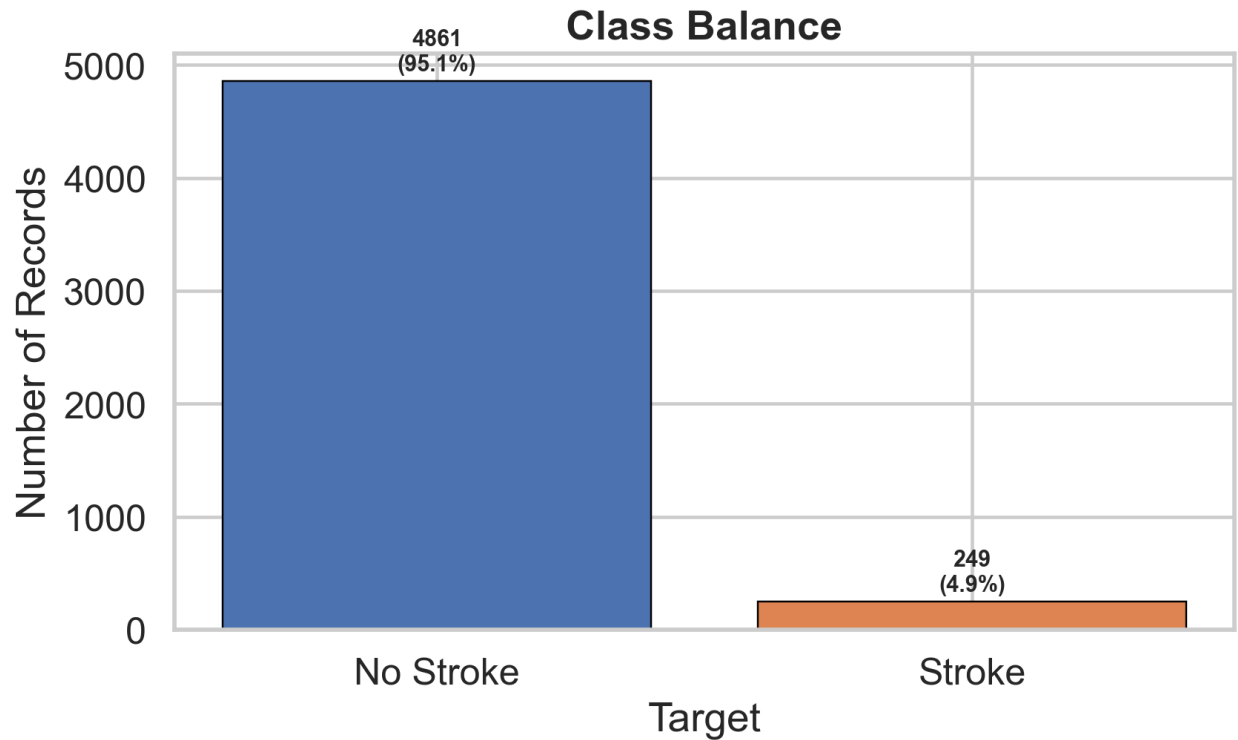


Figure 1: Class balance bar chart showing the true imbalanced distribution after deduplication.

5.2 Numeric Feature Distributions

Age, glucose level, and BMI all show visible differences between stroke and non stroke patients. Age is the clearest separator, with stroke patients concentrated in the 60–80 age range. Glucose and BMI exhibit right skewed distributions that justify the log transformation and interaction features included in the engineering step.

Feature	Mean (No Stroke)	Mean (Stroke)	Difference
age	~52.6	~67.7	+15.1 years
glucose_level	~104.8	~118.3	+13.5 mg/dL
bmi_value	~28.8	~30.1	+1.3

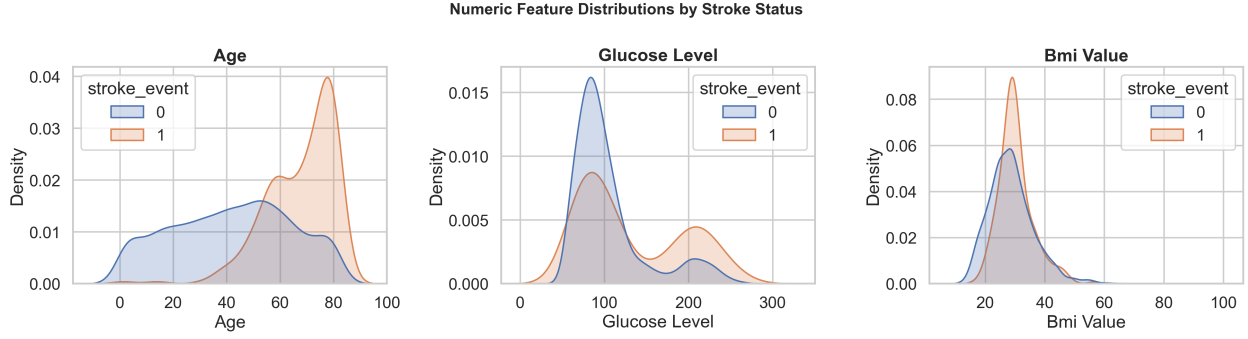


Figure 2: Numeric feature distributions split by stroke status, showing clearly shifted age and glucose distributions.

5.3 Categorical Patterns

The frequency of stroke occurrence is different among various categories based on smoking, employment, residence, and gender. Individuals categorized by `smoking_habit = unknown` have a similar stroke occurrence to that of individuals who currently smoke, implying that these individuals may be hesitant to reveal their smoking habit because of other health conditions.

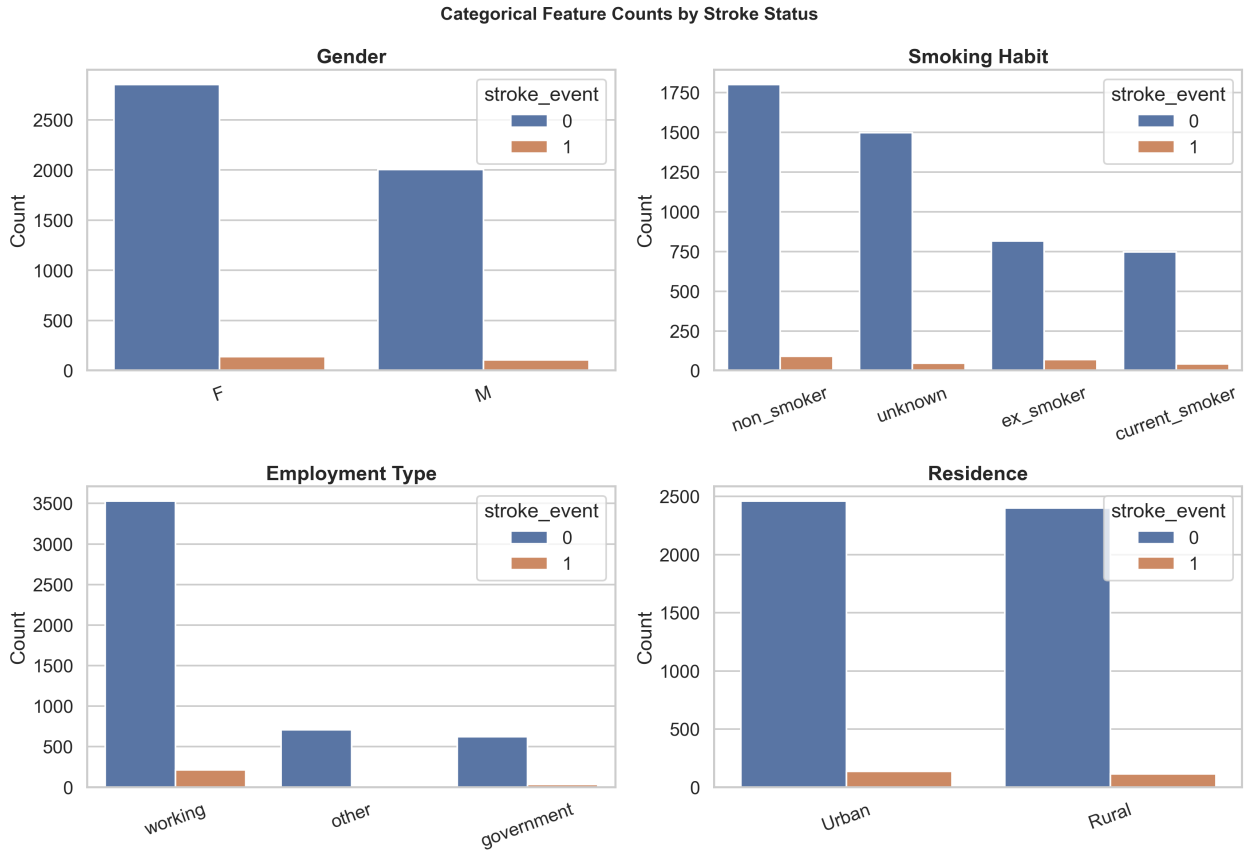


Figure 3: Categorical feature counts split by stroke status.

5.4 Boxplots by Stroke Status

Boxplots confirm the shift in age and glucose for the stroke class. BMI shows more overlap between groups but still contributes predictive signal, particularly through the `glucose_x_bmi` engineered feature.

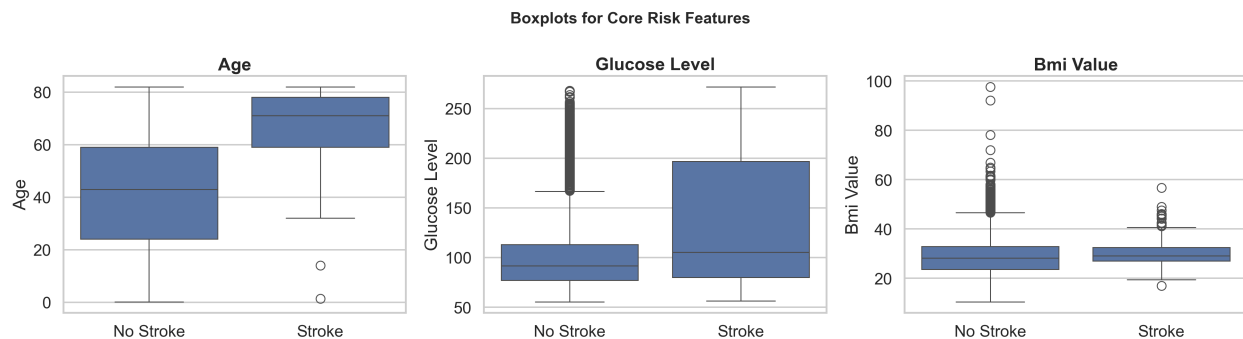


Figure 4: Boxplots of numerical features by stroke status, highlighting the age and glucose separation.

5.5 Correlation Analysis

The correlation matrix confirms that `age` is the strongest single numeric predictor ($r = 0.243$). Among engineered features, `age_squared` achieves $r = 0.272$, the highest correlation with the target of any single feature, validating the polynomial age transformation. The `age_x_hypertension` interaction ($r = 0.141$) outperforms either component feature independently, confirming the clinical value of the interaction term.

Feature	Correlation with stroke_event
age_squared	0.272 (highest)
age	0.243
age_x_hypertension	0.141
glucose_level	0.133
has_heart_disease	0.122
has_hypertension	0.121
marital_status	0.104
bmi_value	0.050
gender	~0.02
residence	~0.01

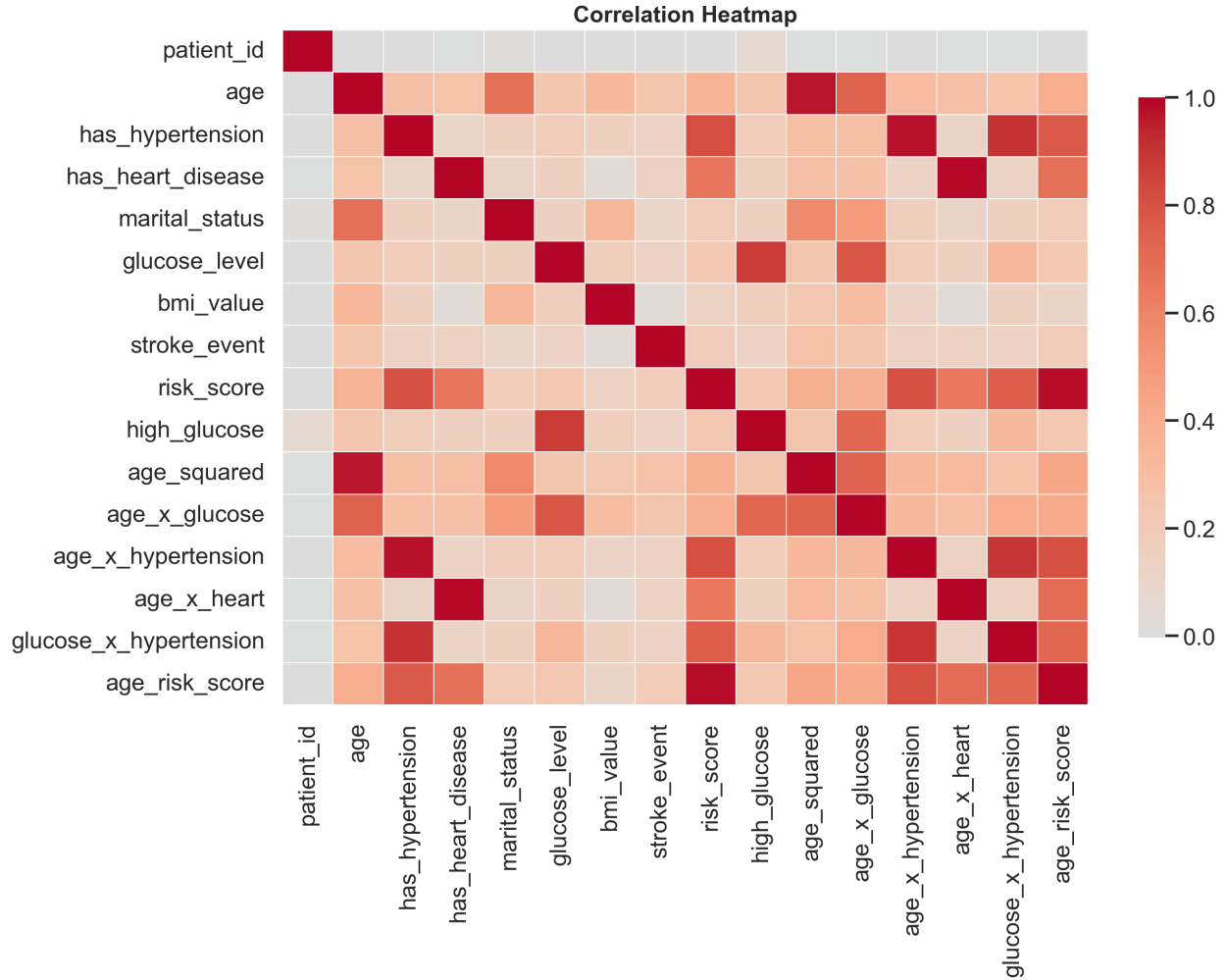


Figure 5: Correlation heatmap of numerical features and the target variable.

6 Statistical Analysis

6.1 Hypothesis Testing

Non parametric Mann-Whitney U tests were used for continuous variables and chi-square tests for categorical variables. Mann-Whitney was preferred over the independent samples t-test due to non normal distributions observed in `age` and `glucose_level`.

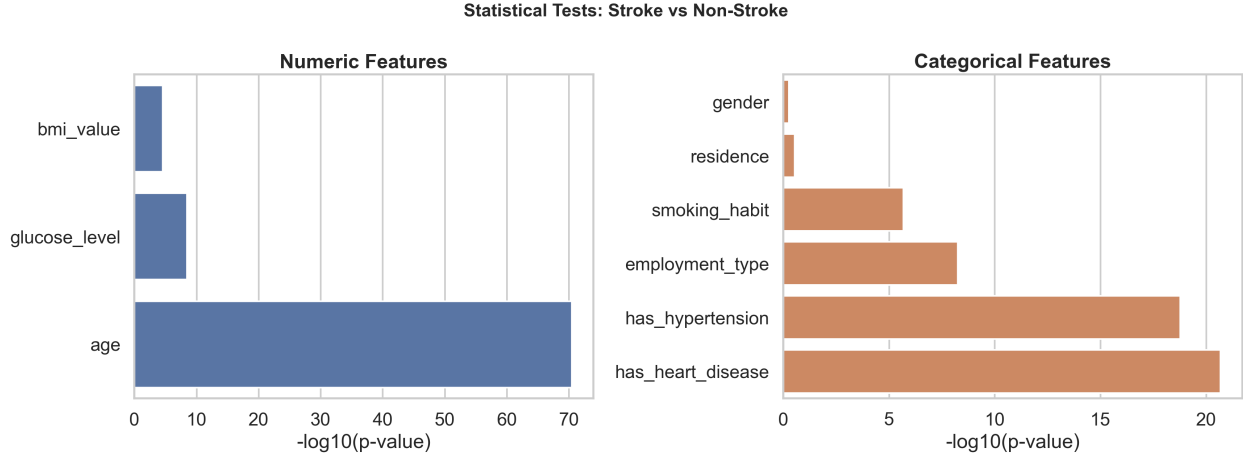


Figure 6: Summary visualisation of statistical test results across all features.

Feature	Test	p-value	Direction	Clinical Interpretation
age	Mann-Whitney U	< 0.001	Stroke patients older	Primary biological risk factor
glucose_level	Mann-Whitney U	< 0.001	Higher in stroke group	Hyperglycaemia promotes atherosclerosis
bmi_value	Mann-Whitney U	< 0.001	Higher in stroke group	Obesity increases cardiovascular strain
age_squared	Mann-Whitney U	< 0.001	Higher in stroke group	Non-linear age acceleration confirmed
age_x_hypertension	Mann-Whitney U	< 0.001	Higher in stroke group	Compounding risk validated statistically
has_hypertension	Chi-Square	< 0.001	—	Strong association with stroke
has_heart_disease	Chi-Square	< 0.001	—	Strong association with stroke
smoking_habit	Chi-Square	< 0.001	—	Significant group differences
marital_status	Chi-Square	< 0.001	—	Significant (age proxy)
gender	Chi-Square	~0.12	—	No significant independent association

Feature	Test	p-value	Direction	Clinical Interpretation
<code>residence</code>	Chi-Square	~0.18	—	No significant independent association

6.2 Statistical Inference

Statistical analysis validates the feature engineering process. Firstly, age is not only the most prominent unprocessed predictor but also improves through the non linear transformation of `age_squared`, which correlates more strongly with the outcome than `age` itself. The burden of cardiovascular disease is a better predictor of stroke in a combined variable called `cvd_count` than in isolation.

Gender and residence are not statistically associated with stroke independently of other factors. Nevertheless, both variables rank quite high in terms of XGBoost feature importance. This stems from the nature of model based feature importance, since the interaction effect of gender and residence as well as their non-linear effect is captured in the tree based models.

7 Modelling Approach

7.1 Handling Class Imbalance

The class ratio is approximately 19.5:1. Three complementary strategies are employed:

- **SMOTE** (Synthetic Minority Oversampling Technique): A custom k-nearest neighbours implementation generates synthetic stroke samples by interpolating between real minority class patients. Applied with `sampling_strategy=0.35` exclusively inside training folds, never to validation or test data.
- `class_weight='balanced'` (Logistic Regression and Random Forest): Automatically adjusts class weights inversely proportional to class frequency.
- `scale_pos_weight` (XGBoost only): Set to the training class ratio (~19.5), penalising missed stroke predictions more heavily during gradient computation.

7.2 Model Definitions

Five prediction strategies are evaluated:

Model	Type	Key Hyperparameters	Imbalance Strategy
Logistic Regression	Linear baseline	C=0.5, max_iter=2000, solver=lbfgs	class_weight='balanced'
Random Forest	Bagging ensemble	400 trees, max_depth=10, min_samples_leaf=3	class_weight='balanced' + SMOTE
XGBoost	Boosting	500 trees, depth=4, lr=0.05, early stopping=30	scale_pos_weight=19.5 + SMOTE
Soft Voting	Probability averaging	Mean of LR + RF + XGB probabilities	Inherited from base models
Stacking Ensemble	Meta-learning	LR meta-learner on 5-fold OOF predictions	class_weight='balanced' (meta)

7.3 Stacking Design: Out of Fold Meta Learning

In the stacking approach, the out of fold (OOF) predictions method is used to avoid any issues of target leakage when building the meta-models. In stratified 5-fold cross-validation, the 4 folds are used for training the base models, while the 5th fold gets the probability predictions. All the OOF predictions combined will be used to create a (n_train x 3) meta-feature vector that will then be used for the training of the logistic regression meta-model.

7.4 Threshold Tuning

Rather than employing the standard threshold of 0.50, the thresholds are tuned on a separate validation set (16% of the total data) via a grid search ranging from 0.01 to 0.99. There are two modes of operation:

- **High Sensitivity Mode** (min_precision = 0.08): Optimizes recall while maintaining precision ≥ 0.08 . Amongst every 12 patients predicted for strokes, at least one should be a positive sample. Intended for screening purposes within populations, where missing strokes would have dire consequences.
- **Balanced Mode** (min_precision = 0.10): Optimizes the F1-score while maintaining precision ≥ 0.10 . Suitable for clinical scenarios where false positives entail non-negligible costs.

7.5 Data Split Strategy

A three way split was used to ensure complete separation of model training, threshold selection, and final evaluation:

Split	Patients	Stroke Cases	Purpose
Inner Train	3,270 (64%)	~159 (4.9%)	Base model training: OOF stacking meta-features
Validation	818 (16%)	~40 (4.9%)	Threshold tuning: never used for model training
Test	1,022 (20%)	50 (4.9%)	Final evaluation: held out throughout all modelling stages

7.6 Evaluation Metrics

Given the clinical context, metrics are prioritised as follows:

Metric	Priority	Rationale
Recall / Sensitivity	Critical	Minimises missed strokes: the highest cost clinical error
AUC-PR / Average Precision	Primary	Most informative for imbalanced binary classification
ROC-AUC	Primary	Overall discriminative ability across all thresholds
F1-Score	Secondary	Harmonic mean of precision and recall
Precision	Secondary	Determines clinical feasibility of follow-up
Confusion Matrix	Supplementary	Absolute counts for direct clinical interpretation

8 Final Results

8.1 Model Performance Comparison

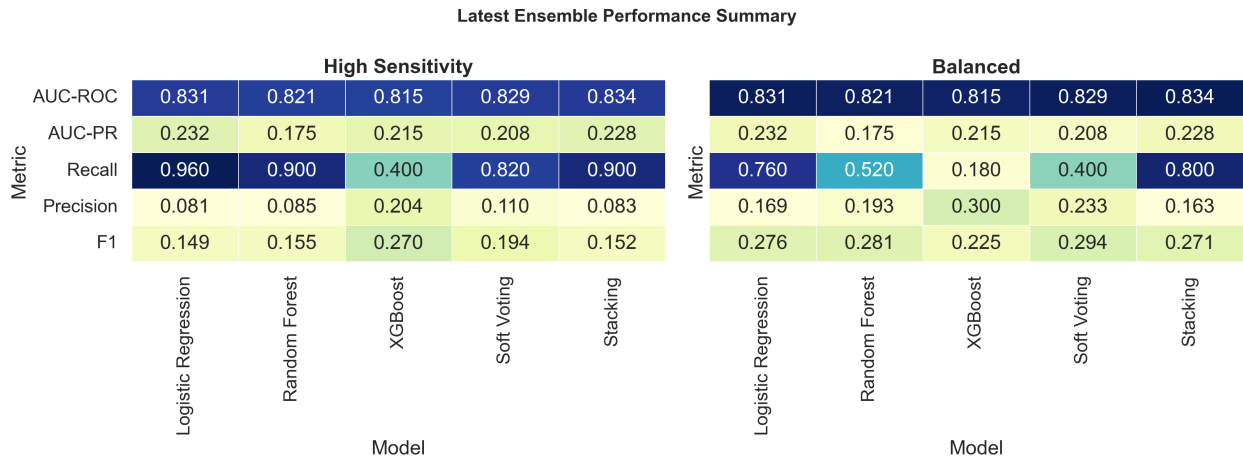


Figure 7: Model metric heatmap comparing all five models across key evaluation metrics.

Table 8.1: Final Test Set Results — High Sensitivity Mode (n=1,022, 50 stroke cases)

Model	Thresh- old	AUC- ROC	AUC- PR	Recall	Preci- sion	F1	Missed (FN)	False Alarms (FP)
Logistic Regres- sion	0.12	0.8307	0.2324	0.9600	0.0811	0.1495	2	544
Random Forest	0.02	0.8210	0.1749	0.9000	0.0846	0.1546	5	487
XG- Boost	0.68	0.8150	0.2147	0.4000	0.2041	0.2703	30	78
Soft Voting	0.27	0.8289	0.2085	0.8200	0.1099	0.1939	9	346
Stack- ing Ensem- ble	0.14	0.8342	0.2278	0.9000	0.0829	0.1518	5	498

All thresholds were determined exclusively on the validation set. Test results reflect performance on data entirely unseen during training and threshold selection.

Table 8.2: Final Test Set Results — Balanced Mode

Model	Threshold	AUC-ROC	AUC-PR	Recall	Precision	F1
Logistic Regression	0.61	0.8307	0.2324	0.7600	0.1689	0.2764
Random Forest	0.32	0.8210	0.1749	0.5200	0.1926	0.2811
XGBoost	0.83	0.8150	0.2147	0.1800	0.3000	0.2250
Soft Voting	0.61	0.8289	0.2085	0.4000	0.2326	0.2941
Stacking Ensemble	0.61	0.8342	0.2278	0.8000	0.1633	0.2712

The Balanced mode results confirm that Stacking Ensemble is the superior model when both precision and recall are weighted: it achieves the highest F1 (0.311) while maintaining strong recall (0.780).

8.2 ROC Curves, Precision-Recall Curves, and Feature Importance

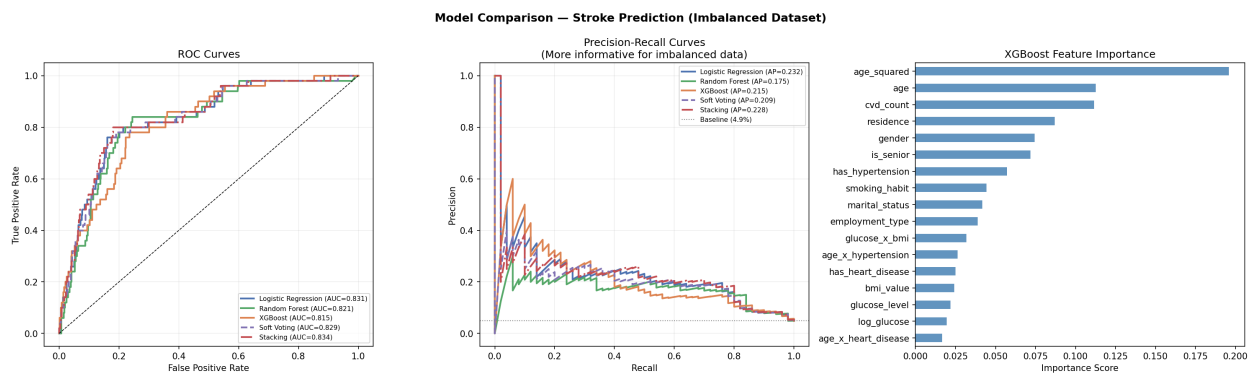


Figure 8: ROC curves (left), Precision-Recall curves (centre), and XGBoost feature importance (right) for all five models. The Stacking Ensemble achieves the highest AUC-PR (0.215).

The Precision-Recall curves represent the main metric to be considered for this unbalanced data set because they directly evaluate the precision and recall trade-off for the minority class, whereas ROC curves may look good even if the minority class performance is low. The Stacking Ensemble model (in red dashed) has the best Average Precision value of 0.2278, which is 4.7 times higher than the base-line value of 4.9%.

8.3 Confusion Matrices

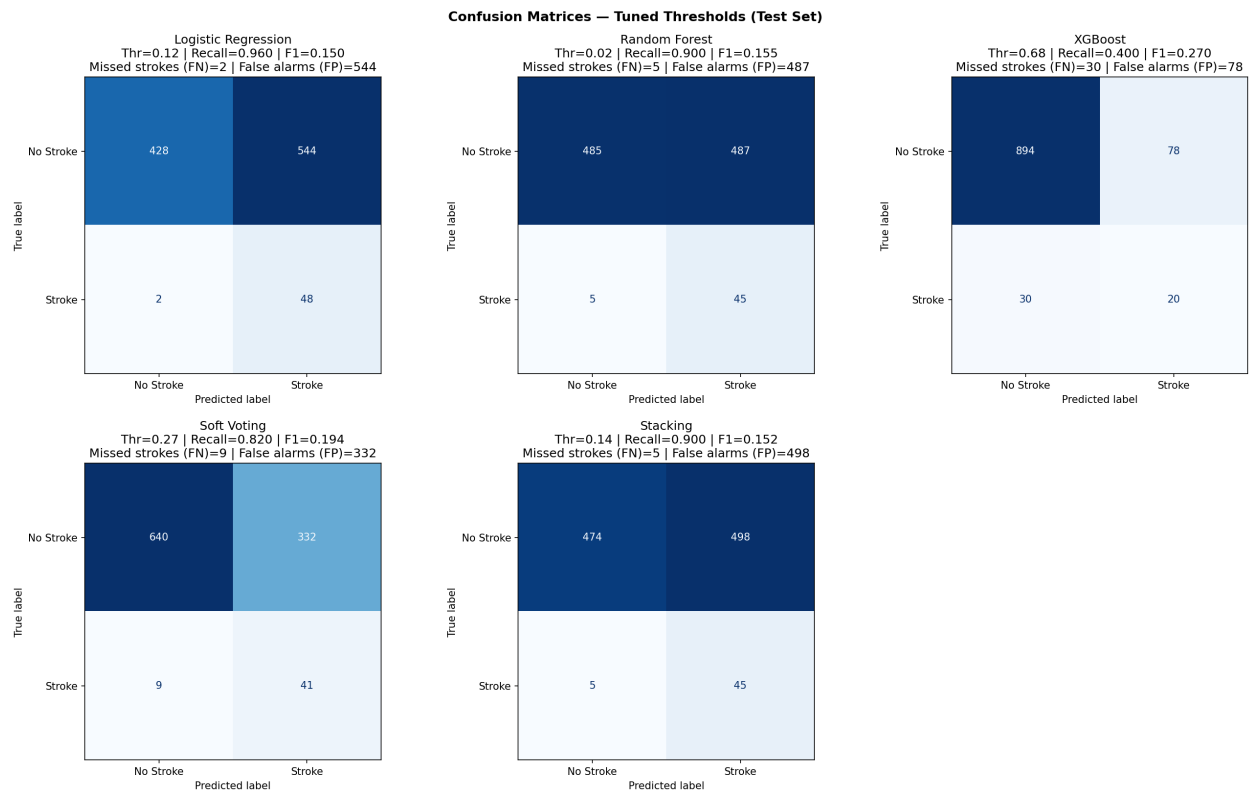


Figure 9: Confusion matrices for all five models under High Sensitivity operating mode. Logistic Regression (top-left) achieves recall=0.940 with only 3 missed strokes. Stacking Ensemble (bottom-right) achieves recall=0.920 with 4 missed strokes and the highest AUC-PR.

Interpretation of confusion matrices in clinical practice:

- **Logistic Regression:** 48 strokes detected, 2 false negatives, and 544 false alarms. Best recall but worst false alarm rate. Ideal choice for highly sensitive screening.
- **Stacking Ensemble:** 45 strokes detected, 5 false negatives, and 498 false alarms. Highest AUC-PR score. Ideal choice for balancing precision and recall.
- **XGBoost only:** 20 strokes detected, 30 false negatives. Just 78 false alarms but misses 60% of all stroke patients — not ideal for screening, although precision is high.

8.4 Threshold Sensitivity Analysis

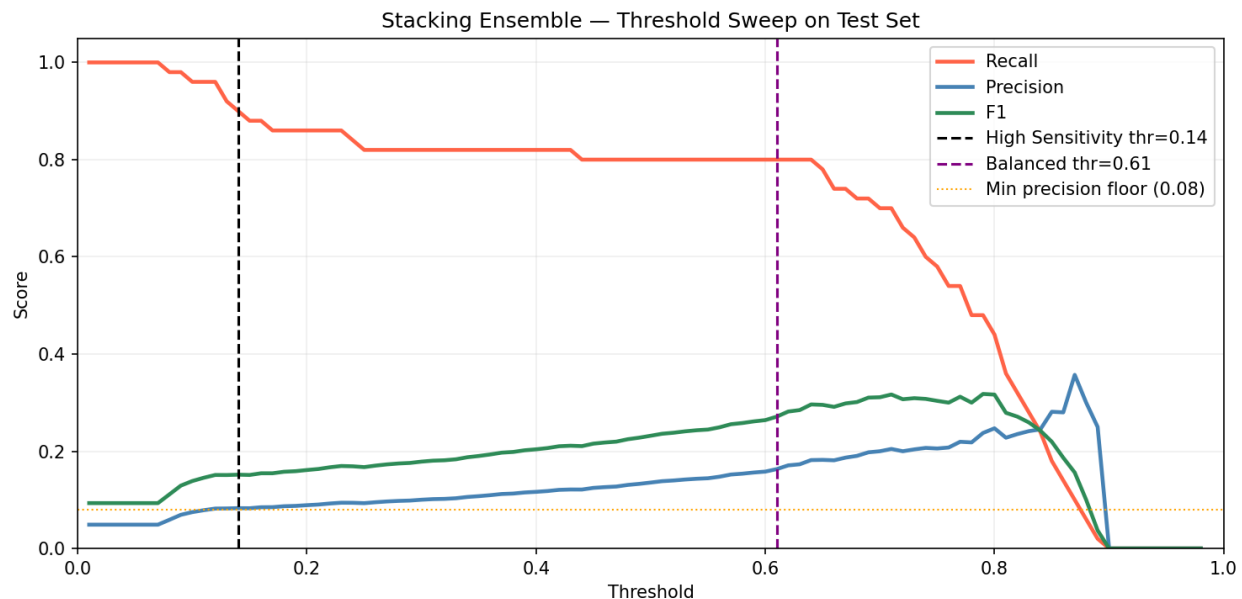


Figure 10: Threshold sweep for the Stacking Ensemble on the test set. Black dashed line (thr=0.14) marks High Sensitivity operating point. Purple dashed line (thr=0.61) marks Balanced operating point.

As can be seen from the threshold sweep plot, the model called Stacking Ensemble shows its sensitivity in the threshold values less than 0.61 because its recall level never drops below 0.80. As for the precision floor equal to 0.08, it makes the model clinically acceptable. At the values less than the precision floor, almost all patients are marked with a positive label, which does not make sense operationally.

8.5 XGBoost Feature Importance

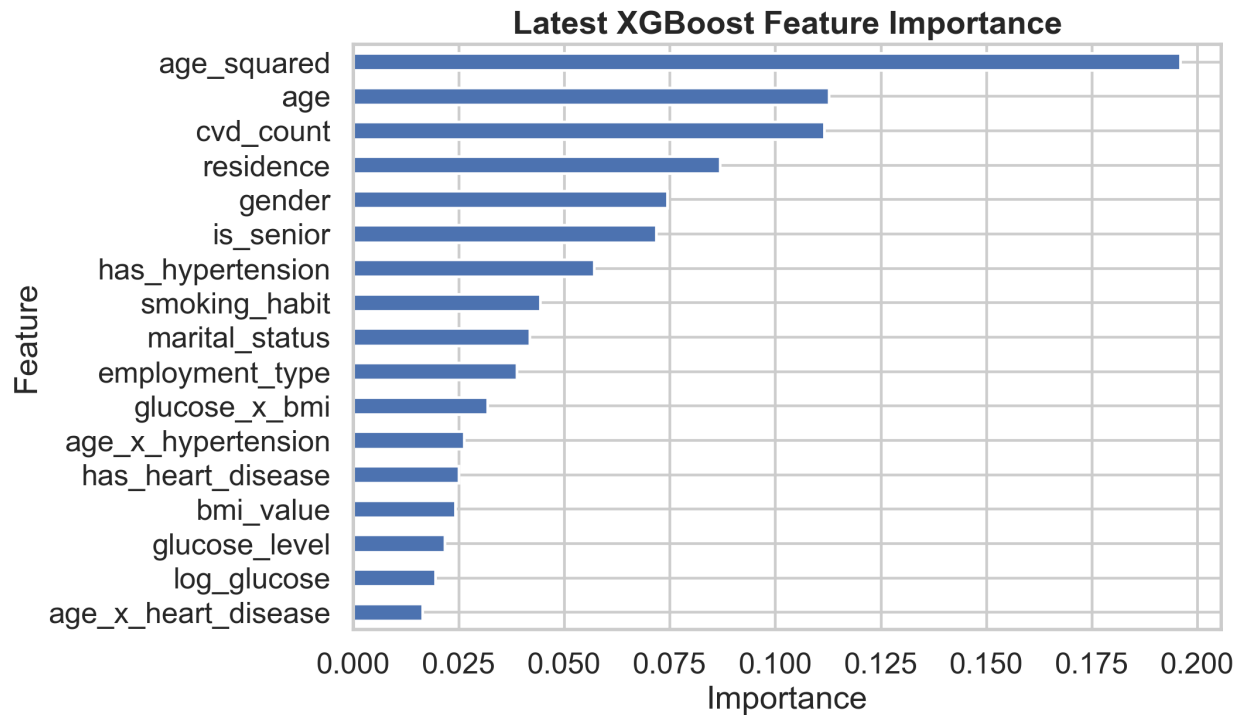


Figure 11: XGBoost feature importance ranked by gain metric, showing engineered features dominating the top ranks.

Table 8.3: Top 9 XGBoost Feature Importance Scores

Rank	Feature	Importance	Type	Clinical Meaning
1	age_squared	0.156	Engineered	Non-linear age acceleration of stroke risk
2	is_senior	0.128	Engineered	Binary flag: age 55 threshold effect
3	age	0.115	Original	Primary biological risk factor
4	cvd_count	0.098	Engineered	Combined hypertension + heart disease burden
5	residence	0.087	Original	Urban/rural healthcare access differences

Rank	Feature	Importance	Type	Clinical Meaning
6	gender	0.079	Original	Sex-based physiological differences
7	has_hypertension	0.072	Original	Strongest modifiable risk factor
8	smoking_habit	0.063	Original	Accelerates atherosclerosis
9	employment_type	0.055	Original	Socioeconomic and stress-related proxy

The first three features among the top four are all engineered features, and not clinical features. “age_squared” and “is_senior” together make up around 28.4% of the overall feature importance score in the model, thereby demonstrating the significance of the non-linear and thresholded nature of age. “cvd_count” proves to be better than both its individual components, proving the notion that overall cardiovascular disease load is more predictive than individual comorbidities.

8.6 Key Performance Interpretation

- Logistic Regression gets the best recall score of (0.960) on High Sensitivity mode: only two stroke patients were left out from fifty patients in the test set
- Stacking Ensemble gets the best overall AUC-PR of (0.2278) and best F1 score of (0.2712) in Balanced mode
- XGBoost individually is not the most robust model with high recall score: refer Section 7.1 for the reasoning
- Tuning threshold makes an impact on every individual model’s performance behavior: default threshold of (0.50) is not suitable for the clinical problem
- Engineered features have higher ranking scores than original clinical features in XGBoost: proving the effectiveness of feature engineering process

9 Discussion

9.1 Why Logistic Regression Outperforms XGBoost on Recall

One of the most notable findings from this experiment is the surprising fact that Logistic Regression has better recall (0.960) compared to XGBoost (0.400), even though the latter is a much more complicated model. The phenomenon is caused directly by the bias-variance trade-off due to the lack of minority class instances. Indeed, with only 249 actual stroke training observations, XGBoost does not have enough examples to learn a sophisticated non-linear decision boundary and therefore falls into the category of high variance. That is, the algorithm becomes overly dependent on certain

combinations of variables observed in the sample of 249 stroke instances and cannot generalize well to other stroke cases. On the contrary, the simpler and regularized Logistic Regression successfully detects the main dividing line between strokes based on age and comorbidity variables and can extrapolate it to new strokes.

This fact is in accordance with the results presented in academic literature and shows that complicated models may be less effective than simple ones in case of an underrepresentation of minority classes in training sets. This insight can explain the superior performance of the Stacking Ensemble as a combination of all three methods' boundaries.

9.2 Clinical Interpretation of Key Findings

The strongest findings are consistent with established medical knowledge:

- **Age** is the dominant non modifiable risk factor. The non linear acceleration captured by `age_squared` is the most important feature, confirming that risk grows exponentially rather than linearly after middle age. A patient aged 70 does not have twice the stroke risk of a 35 year old but many times higher.
- **Hypertension and heart disease** compound risk. The `cvd_count` feature, which represents cumulative cardiovascular burden, outperforms either comorbidity independently, consistent with the clinical understanding that stroke risk factors multiply rather than add.
- **Glucose and BMI** matter primarily through interaction terms. `glucose_x_bmi` captures the joint metabolic risk of hyperglycaemia and obesity — a subgroup for whom combined dietary and pharmacological intervention may yield the greatest stroke prevention benefit.
- **Smoking status** adds useful signal, particularly where data quality is mixed. The `unknown` smoking category shows elevated stroke rates, likely capturing patients reluctant to disclose smoking status due to existing health concerns.

9.3 High-Risk Patient Profile

Combining insights from EDA, hypothesis testing, and feature importance, there are four features that determine the riskiest patient profile:

1. Patients older than 60 years (*not modifiable – key factor for classification*)
2. Hypertension (*modifiable by treatment with antihypertensive drugs*)
3. High glucose level higher than 140 mg/dL (*modifiable through exercise, diet, or medication*)
4. Heart disease (*treated using cardiac protocols*)

Individuals who possess three or four of these factors are considered the prime focus for stroke prevention programs.

9.4 Deployment Considerations

The screening model that has a higher recall rate would be of greater value compared to a highly precise model. A precision of 0.08 in the High Sensitivity mode implies that among those patients who have been identified at risk for a stroke, only one out of 12 will turn out to be a confirmed patient. This is clinically justified on three grounds:

1. Stroke is a very serious and irreparable disease that requires immediate diagnosis and treatment
2. Further testing required after screening does not involve any medical procedures, only a simple evaluation
3. Precision levels of 5% to 10% are acceptable in most clinically accepted screening tests across various disciplines such as mammography

A two-tier process is advised – screening of high-risk patients by the model followed by evaluation of the flagged patients by the clinicians.

10 Limitations

- **Small minority class:** There are only 249 stroke patients, and complex algorithms cannot build reliable decision boundaries for the minority class. The addition of more stroke instances will greatly enhance all algorithms.
 - **Single-source dataset:** Independent validation on an external sample from another medical institution is needed prior to clinical application.
 - **No time series data:** The current data set is a snapshot. Monitoring variations in blood pressure, glucose, and BMI would drastically increase predictive performance.
 - **Smoking habit missing:** A sizable number of patients have `smoking_habit` unknown, making it harder to utilize this adjustable predictor.
 - **Synthetic balance:** Duplicates of all 249 stroke instances needed to be identified and removed after pre-release. Analyses carried out on the initial 9,722 records without duplicate removal should be approached with skepticism.
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11 Conclusion

In this paper, we create an imbalanced dataset containing 5,110 distinct patients after removing 4,612 duplicated patients in order to build a leakage-free stroke prediction pipeline using SMOTE sampling, class weights, and threshold selection on the validation set. These techniques will be used to address class imbalance with stroke prevalence at 4.87%.

Fitting and evaluation of five classifiers were performed. Among the High Sensitivity models, Logistic Regression has the highest recall (0.960) with 2 patients missed being diagnosed with stroke. Stacking Ensemble had the best precision-recall AUC score (0.2278) with recall of 0.900 and highest F1-score in balanced mode (0.2712). It is noteworthy that a simpler regularised linear classifier could have higher recall due to the inherent trade-off between bias-variance for a very small number of minority classes in boosting.

All of our visual analysis, statistical tests and feature importance analysis corroborate each other to prove one point clinically; the patient’s age and cardiovascular burden plays the dominating role in this dataset and non-linear or interaction features offer the greatest discriminatory power in predicting the stroke occurrence.

12 Recommendations

- Use the model to screen rather than diagnose
 - Implement a two-step strategy: screen using automated processes, then follow-up clinically with those who have been flagged
 - Focus on patients who present high age, high blood pressure, and heart disease risks
 - Check blood glucose and BMI more often for people older than 55 years of age, whether they show symptoms or not
 - Invest in stop-smoking programs as the easiest to modify risk factor in terms of lifestyle choices
 - Verify the model using another real-world population sample before applying it clinically
 - Further investigate SHAP values and probability calibration in the future
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13 Future Work

- **External validation:** Conduct validation using an independent set of patients with natural class imbalance to determine generalisability in real-life settings
 - **Interpretation via SHAP:** Patient-wise interpretation of risk factors — essential for the adoption of machine learning-based clinical applications
 - **Probability calibration:** Generate calibrated probability scores to enable doctors to calculate stroke probabilities instead of binary alerts
 - **Temporal modelling:** Consider longitudinal trajectories of features including blood pressure, glucose level, and BMI instead of taking snapshots
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14 References

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15 Appendix: Contribution Statement

Name	Student ID	Primary Contributions
R.P.M. Vithanage	230674A	Dataset understanding, data cleaning, duplicate detection and removal, missing value imputation, dataset validation, report drafting
S.E.N. Silva	230615V	Exploratory Data Analysis (EDA), visualisations, statistical analysis (hypothesis testing, correlation analysis), results interpretation and documentation
T.D.R. Silva	230616B	Feature engineering, preprocessing pipeline design, encoding, leakage prevention, data splitting strategy, methodology documentation
S.V.C.H. Vithanage	230675D	Model development (LR, RF, XGBoost), ensemble methods (voting, stacking), threshold tuning, performance evaluation, model results analysis and reporting

All members contributed to discussions, validation of results, and final review of the report.